PAULSON ALEX FERNANDES

Software Developer | Cybersecurity Analyst

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in **Cybersecurity and Digital Forensics**, with hands-on experience in software development and security analysis. Proficient in **Python, JavaScript, C++, and SQL**, with a strong foundation in Object-Oriented Programming, Data Structures, and Algorithms. Developed practical projects in malware detection, network security, digital forensics, and database management, eager to apply problem-solving and technical skills to build reliable and impactful solutions across technology domains.

WORK EXPERIENCES

Cybersecurity Intern | Internship Aug 2025 - Aug 2025

Btech Walleh *Remote*

- Investigated **Conti ransomware** in Splunk, identifying persistence mechanisms, credential-theft indicators, and exploited CVEs.
- Analyzed **Lumma Stealer** activity using Wireshark and Zeek to identify malicious domains, C2 traffic, and data-exfiltration indicators.
- Conducted controlled Windows penetration tests using **Metasploit** and **EternalBlue**, including privilege escalation and password-hash cracking.
- Simulated ARP poisoning and configured Windows Firewall rules to demonstrate network risks and strengthen endpoint defenses.

Technologies / Skills Used : *Splunk, Wireshark, Zeek, Metasploit*

PROJECTS

AI Malware Detection System | Github

Developer

- Built a static-analysis pipeline to classify Windows executables using 50+ structural, entropy, and import/export features based on **Data Structures** and **Algorithms**.
- Developed model comparison, explainability, scan-history tracking, and downloadable PDF security reports. Additionally, designed the system architecture to ensure **Distributed Systems** principles, focusing on **Application Scalability** and **Cloud environments**.

Technologies / Tools Used : *Python, Flask, Random Forest, XGBoost, SQLite, ReportLab, Data Structures, Algorithms, Distributed Systems, Application Scalability, Cloud environments*

HoneyWatch – SSH Threat Intelligence Honeypot | Github

Developer

- Designed an SSH honeypot with stateful fake-shell emulation and canary credentials to capture attacker behavior and malware samples.
- Implemented GeoIP enrichment, attack-pattern detection, and real-time WebSocket monitoring of attacker origins.

Technologies / Tools Used : *Python, Paramiko, asyncio, SQLite, WebSockets, GeoIP*

VaultIX – Secure Password Manager | Github

Developer

- Developed a browser-based password vault with client-side AES-256-GCM encryption and PBKDF2 key derivation.
- Implemented cryptographically random password generation, entropy-based strength scoring, auto-lock, and secure import/export.

Technologies / Tools Used : *JavaScript, Web Crypto API, AES-256-GCM, PBKDF2*

FORGE-X | Digital Forensics Evidence Management System | Github

Developer

- Developed a web-based digital forensics management system using **Flask and MySQL** to manage investigation cases, digital evidence, examinations, and forensic reports.
- Implemented **SHA-256 evidence integrity verification**, chain-of-custody tracking, role-based access control, and audit logging to support evidence traceability.
- Designed a normalized relational database with constraints, views, stored procedures, triggers, and transaction management to enforce data integrity and business rules.

Technologies / Tools Used : *Python, Flask, MySQL, HTML, CSS, JavaScript*

IPsec VPN Analyzer Dashboard | Github

Developer

- Developed a **cybersecurity analysis platform** to inspect IPsec VPN packet captures, extract IKE negotiation parameters, and assess cryptographic configuration security.
- Implemented rule-based security scoring, threat findings, and metadata exposure analysis to identify risks observable in encrypted traffic without decrypting payloads.
- Integrated a **Random Forest classifier** to analyze ESP traffic patterns using packet sizes, timing, throughput, and flow statistics; built interactive dashboards and technical security reports.

Technologies / Tools Used : *Python, FastAPI, React, Scikit-learn, Random Forest, Scapy, Wireshark, SQLite*

SKILLS

Databases :	MySQL, SQLite
Frameworks & Libraries :	Scikit-learn, XGBoost, Random Forest
Programming Languages :	Python, C++, JavaScript, C, Object-Oriented Programming, AI coding assistants, Claude, GitHub Copilot
Soft Skills :	Communication, Teamwork, Leadership, Problem Solving, Decision Making
Tools & Platforms :	Splunk, Wireshark, Zeek, Metasploit, Nmap, Microsoft Sentinel, Microsoft Defender, Ettercap, John the Ripper, Linux

EDUCATIONS

MIT World Peace University | Pune 2025 - 2029

B.Tech, CSE (Cybersecurity and Forensics)

CERTIFICATIONS

- **HackerRank**: Software Engineer Intern (Sep 2026)
- **Palo Alto Networks Academy**: Cybersecurity Fundamentals (Nov 2025)
- **Palo Alto Networks Academy**: Network Security Fundamentals (Apr 2026)
- **Launched Global**: Cyber Security Mentorship Program (Certificate of Course Completion)

LEADERSHIP & LANGUAGES

Core Team Member August 2026
Smart India Hackathon 2025 (Internal)
- Served as a Core Team Member, coordinating technical operations and managing events.
- Facilitated team coordination, evaluations, and the execution of events.

Student Co-ordinator March 2026
CIPHATHON 26
- Assisted in event planning, coordination, and technical operations for CIPHATHON 26.
- Facilitated smooth execution and collaboration among participants and the organizing team.

Languages
- English (**Full Professional**)
- Hindi (**Native**)
- Konkani (**Native**)